

AN AGENTIC AI ARCHITECTURE FOR NATIONAL-SCALE FRAUD DETECTION IN DIGITAL PAYMENTS ECOSYSTEMS

A HIGH-LEVEL DESIGN PERSPECTIVE

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ABSTRACT

Real-time national fraud detection platforms — exemplified by Brazil's PIX, India's UPI and Nigeria's NIBSS — embed centralised AI-and-rule-driven monitoring at the settlement layer, yet share three structural limitations: human-in-the-loop investigation bottlenecks, batch retraining latency and weak cross-institution coordination. This paper proposes an architectural successor: an agentic AI fabric in which autonomous agents, with bounded autonomy, persistent memory and observable handoffs, distribute detection, investigation, remediation and learning across the terminal, institutional and national-authority layers. A worked example anchored on the NIBSS National Payment Stack in Nigeria, under the Central Bank of Nigeria's January 2025 fraud-debit directive, illustrates the design end-to-end. Extensibility to AML, KYC/KYB, consumer protection and cross-border supervision is discussed, with the principal risks of model governance, accountability and regulatory acceptance. The proposition aligns with the BIS Innovation Hub's emerging SupTech direction.

KEYWORDS

Agentic AI, National Payment Systems, Fraud Detection, SupTech, Embedded Systems for Finance

1. INTRODUCTION

Real-time digital payment systems now process tens of billions of transactions per month at national scale. Brazil's PIX handles more than three billion transactions monthly with sub-0.01% fraud rates; India's Unified Payments Interface (UPI) clears over fifteen billion monthly transactions; and Nigeria's NIBSS Instant Payment (NIP) underpins the country's cashless economy with 24/7 inter-bank settlement [1][2][3]. Each of these systems pairs instant settlement with a centralised real-time fraud detection layer — Brazil's SPI/DICT, India's NPCI Fraud Risk Monitoring (FRM), and the NIBSS-operated fraud monitoring service — combining machine learning models, rule engines and human-led investigation workflows.

These platforms represent a generational achievement in payment infrastructure. They are also reaching the limits of their original architectural paradigm. The June 2025 cyberattack on C&M Software, in which more than one billion reais was siphoned from Central Bank of Brazil reserve accounts through cryptographically valid PIX transactions, exposed how centralised monitoring can be evaded when third-party integrations lack contextual behavioural detection [4]. India's

NPCI is rolling out predictive fraud alerts and tightening velocity controls in response to a corresponding rise in social-engineering scams [5]. In Nigeria, the Central Bank of Nigeria (CBN) issued a January 2025 directive instructing NIBSS to debit the settlement accounts of banks that receive fraud proceeds, signalling a fundamental shift in accountability allocation across the ecosystem [6].

The common thread across these developments is that fraud is no longer a transaction-level decision problem. It is a multi-step workflow that spans investigation, remediation, cross-institution coordination and continuous adaptation — and the workflow is currently bottlenecked at exactly the points where speed matters most. This paper argues that the next architectural generation of national fraud detection should be designed not as a centralised platform with human analysts closing the loop, but as an agentic AI fabric in which specialised, bounded-autonomy agents operate at each ecosystem layer, share a persistent memory substrate, and continuously close the learning loop without batch retraining delays.

The contribution of this paper is threefold. First, it identifies three structural limitations shared by first-generation national fraud platforms. Second, it proposes a layered agentic AI fabric architecture with explicit agent role taxonomy, memory substrate, orchestration patterns and bounded autonomy boundaries. Third, it walks through a worked example at the NIBSS / National Payment Stack layer in Nigeria, anchored in the CBN's January 2025 policy direction. The paper closes with a discussion of extensibility beyond fraud, the principal risks, and open questions for further research.

2. BACKGROUND AND RELATED WORK

2.1. First-generation national fraud detection platforms

Brazil's PIX, launched by the Central Bank of Brazil in November 2020, routes every transaction through the SPI (Sistema de Pagamentos Instantâneos) settlement infrastructure and the DICT alias directory, both operated by the central bank [1][7]. Each participating institution layers its own fraud analytics on top of national monitoring, and BCB Resolution 589 mandates a Special Refund Mechanism (MED) by which fraud victims can request return of funds, with blocking and seven-day evaluation windows enforced across institutions [8]. Updates issued in late 2025 introduced precautionary holds and device-trust-based transfer limits, reflecting an iterative tightening of the architecture without disruption to standard user flows.

India's Unified Payments Interface, operated by the National Payments Corporation of India (NPCI), has implemented a real-time fraud risk monitoring service (FRM) using AI and machine learning, made available to participating banks and payment service providers as a value-added service [2]. NPCI's central switch performs validation and routing for billions of monthly transactions, with PSPs running parallel ML models that score transactions against multiple parameters in real time. The 2025 generation of NPCI alerts introduces predictive fraud detection that warns users of risky payments before confirmation [5].

Nigeria's NIBSS operates the NIBSS Instant Payment (NIP) platform, the central switch through which inter-bank transfers are validated, routed and settled, with end-of-day settlement at the Central Bank of Nigeria. The newly launched National Payment Stack (NPS) extends NIP with native ISO 20022 messaging, multi-identifier tracking (BVN, TIN, RC No.), built-in fraud detection and cross-border support [3]. NIBSS has licensed fraud monitoring services for member

banks. The January 2025 CBN directive instructing NIBSS to debit settlement accounts of banks receiving fraud proceeds represents a structural change in the locus of accountability: institutions are now penalised for failures of post-receipt detection, not only for failures of pre-authorisation screening [6].

2.2. SupTech, RegTech and agentic AI in supervision

The Bank for International Settlements Innovation Hub has invested heavily in AI-assisted supervisory technology. Project Aurora, conducted with the Nordic Centre and Lucinity, demonstrated the superiority of AI-driven typology detection over conventional rule-based monitoring in cross-border anti-money-laundering scenarios, using synthetic data to simulate complex laundering activities and emphasising the value of cross-institution and cross-border cooperation [9][10]. Project Noor, launched in August 2025 with the Hong Kong Monetary Authority and the UK Financial Conduct Authority, applies explainable-AI techniques to financial supervision, recognising that high-risk AI used in fraud screening, credit and onboarding must be interpretable and auditable [11]. The Cambridge SupTech Lab's 2024 benchmarks report that seventy-one percent of financial authorities are now engaged in some form of SupTech activity, with maturity ranging from manual to tech-enabled to AI-driven [12].

In parallel, agentic AI patterns have emerged in commercial financial services. AI agents now perform end-to-end work — including investigation case construction, regulatory report generation, customer outreach and remediation orchestration — within bounded operational policies rather than answering single prompts in isolation. The shift from prompt-and-response AI to agentic AI is increasingly visible in production fintech deployments. This paper proposes that the same operating model applies, at the architectural level, to national fraud detection.

2.3. The gap addressed by this paper

Existing academic and institutional literature treats centralised national fraud detection and emerging agentic AI as separate streams. National platform research focuses on volume, latency, model accuracy and architectural resilience [13][14]. SupTech literature focuses on individual supervisory use cases and explainability [11][12]. There is currently no published high-level design proposition that fuses the two: an architectural model that reframes the entire national fraud detection workflow as an agentic AI fabric spanning terminal, institutional and authority layers. This paper closes that gap.

3. STRUCTURAL LIMITATIONS OF FIRST-GENERATION PLATFORMS

First-generation national fraud detection platforms share three structural limitations that constrain their effectiveness at current and projected transaction volumes.

3.1. The human-in-the-loop investigation bottleneck

Centralised platforms generate alerts in real time but route investigation to human analysts in serial workflows. At national scale, the alert volume far exceeds analyst capacity. The result is either alert suppression — quietly relaxing thresholds to reduce volume, which depresses detection — or alert backlog, in which actionable signals age beyond the window in which they can be acted upon. In real-time payments, this window is measured in seconds for blocking and minutes for recall, after which funds are typically beyond reach. Brazil's experience with the

C&M attack, where one billion reais moved through mule and crypto-exchange accounts within hours, illustrates the cost of investigation latency at the network level [4].

3.2. Batch retraining latency

ML models in current national fraud platforms typically retrain on weekly or monthly cadences. New fraud typologies — particularly social engineering, authorised push payment fraud, and AI-amplified scams [15] — emerge and propagate faster than the retraining cycle. Between cycles, the model is structurally blind to patterns it has not yet learned, and rule-engine patches are imperfect substitutes. The retraining cycle assumes a relatively stable adversary; the contemporary fraud adversary is anything but stable.

3.3. Weak cross-institution coordination

When fraud spans multiple banks — a mule network, a fan-in pattern, a coordinated structuring scheme — current architectures detect each transaction in isolation at each participating bank, with cross-institution pattern detection occurring only at the centre and only after the fact. The directive issued by the CBN in January 2025, holding banks accountable for receiving fraud proceeds, surfaces the structural mismatch: institutions are now liable for cross-institution patterns they cannot independently see. The architectural response should be a coordination layer at the centre that operates in the same real-time tempo as transactions themselves, not a quarterly reconciliation.

These three limitations are not artefacts of insufficient compute or inadequate models. They are properties of an architectural paradigm in which the system is centralised, the workflow is split between machine detection and human action, and the learning cycle is exogenous to the operational cycle. An agentic AI fabric addresses all three simultaneously by distributing the workflow, embedding remediation autonomy within bounded policy, and closing the learning loop continuously.

4. PROPOSED ARCHITECTURE: AN AGENTIC AI FABRIC FOR NATIONAL PAYMENTS

The proposed architecture organises fraud-related capabilities as a layered fabric of cooperating agents. Three operational layers — edge, institutional, and national authority — are underpinned by a cross-cutting governance plane and a foundation of payment standards and sovereign security primitives (Figure 1).

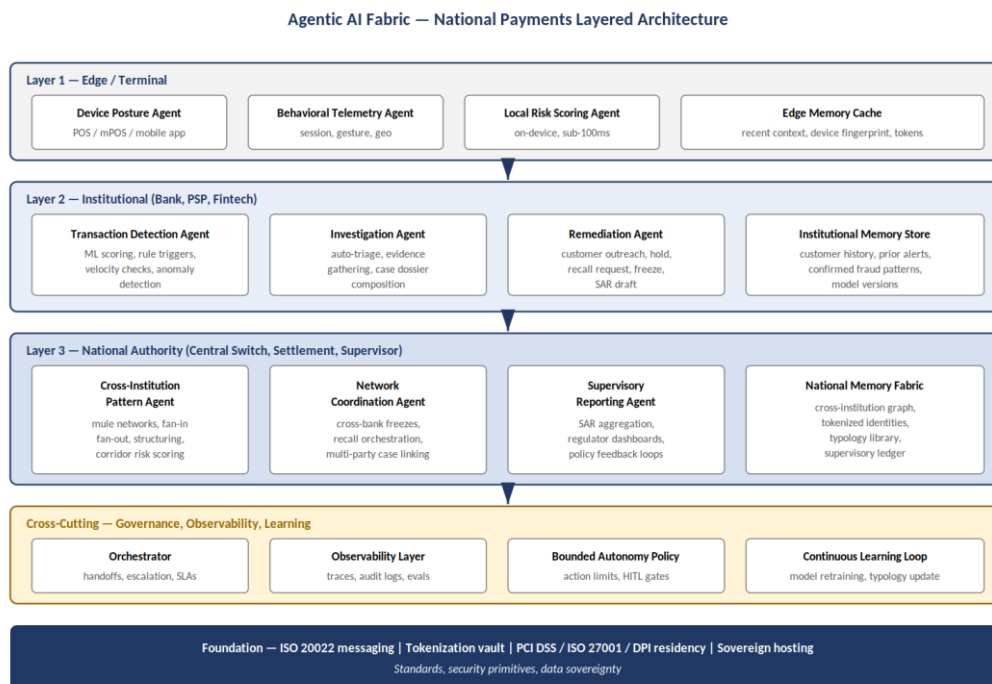


Figure 1 — Agentic AI fabric architecture for national-scale fraud detection: three operational layers underpinned by cross-cutting governance and a foundation of payment standards and sovereign security primitives.

Figure 1. Agentic AI fabric architecture for national-scale fraud detection.

4.1. Operational layers

Layer 1 — Edge / Terminal. Agents operate at the point of customer interaction. A Device Posture Agent verifies device integrity (root, emulator, accessibility-service abuse, overlay risk) drawing on the kind of mobile-banking malware patterns documented in recent threat reports [16]. A Behavioural Telemetry Agent captures session, gesture and geolocation signals. A Local Risk Scoring Agent runs a lightweight on-device model to compute an edge risk score in sub-100ms, attaching reason codes to the outbound ISO 20022 message. An Edge Memory Cache holds short-window context — device fingerprint, recent activity, session tokens — sufficient for the edge agents to operate during brief connectivity losses, which is operationally significant in agency-banking and PTSP environments.

Layer 2 — Institutional. Each participating bank, PSP or fintech runs a quartet of agents on its own infrastructure. A Transaction Detection Agent combines ML scoring with rule triggers, velocity checks and anomaly detection. An Investigation Agent auto-triages alerts, gathers evidence from internal systems of record, and composes case dossiers in machine-readable form. A Remediation Agent executes bounded actions — step-up authentication, precautionary hold, block, recall request, SAR draft — within an explicit policy envelope. An Institutional Memory Store retains customer history, prior alerts, confirmed fraud patterns and model versions. The institutional layer is where most agentic activity occurs; the centre coordinates rather than executes.

Layer 3 — National Authority. At the central switch / settlement / supervisory authority, four agents operate in real time. A Cross-Institution Pattern Agent runs continuous graph analytics over the National Memory Fabric, detecting mule networks, fan-in / fan-out patterns, structuring schemes and corridor risk concentrations that no single institution can see. A Network

Coordination Agent orchestrates cross-bank actions — coordinated freezes, recall requests across multiple legs of a payment chain, multi-party case linking. A Supervisory Reporting Agent aggregates SAR / STR submissions, populates regulator dashboards and produces policy feedback. The National Memory Fabric is a tokenised, graph-structured ledger of cross-institution activity, retaining a typology library curated from observed patterns. This layer is what transforms the architecture from a federation of institutional agents into a coordinated national fabric.

4.2. Persistent memory substrate

Each layer's agents share a memory substrate at that layer (edge cache, institutional store, national fabric). The substrate is not a passive database; it is queried, updated and curated by agents themselves. A confirmed-fraud outcome at the institutional layer triggers updates at the national fabric, which in turn make new typologies available to detection agents at other institutions within minutes rather than the weeks typical of batch model refresh. Persistent memory is what enables agents to behave with continuity across runs rather than as stateless single-call functions, and is the architectural primitive that distinguishes agentic AI from conventional ML services.

4.3. Agent role taxonomy

Across the three layers, agent roles fall into four functional groups: detection, investigation, remediation and learning (Figure 2). Each group encompasses several agent types, and agents within a group cooperate through shared memory and explicit handoff contracts.

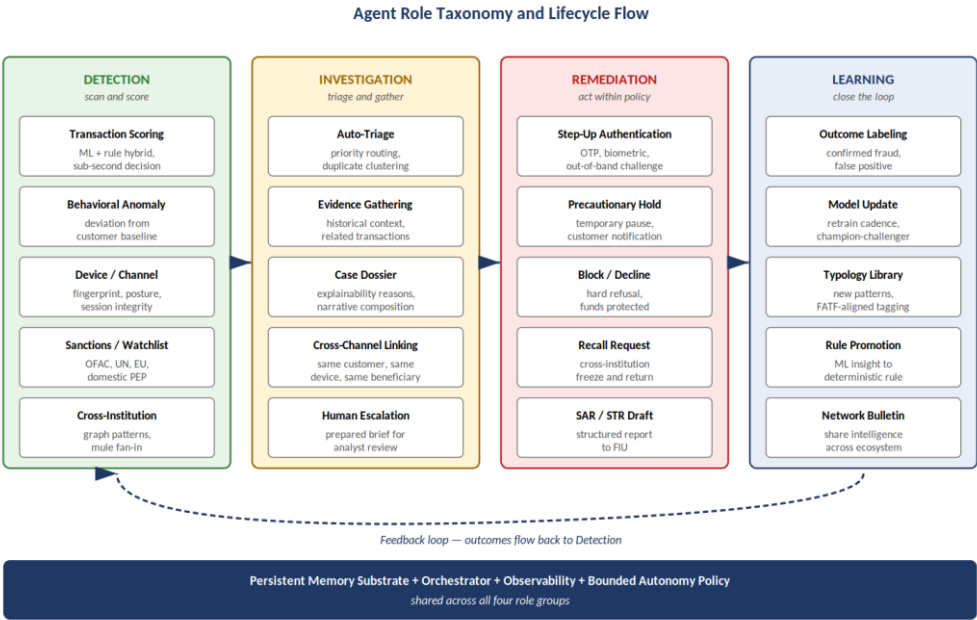


Figure 2 — Agent role taxonomy: four functional groups (Detection, Investigation, Remediation, Learning) operating on a shared memory substrate, with outcomes flowing continuously back into the detection layer.

Figure 2. Agent role taxonomy and lifecycle flow.

The detection group scans incoming activity and produces risk signals. The investigation group consumes those signals, gathers context and constructs the case. The remediation group acts within a bounded policy envelope — step-up, hold, block, recall, report. The learning group ingests outcomes, retrains models, updates the typology library and broadcasts intelligence across the ecosystem. The arrows in Figure 2 show the forward flow; the dashed arrow shows the

feedback loop, which is the architectural commitment that distinguishes this approach from one-shot ML services.

4.4. Bounded autonomy and human-in-the-loop gates

Agentic autonomy is bounded by an explicit policy that defines what each agent class may do without human review and what requires escalation. A Remediation Agent may issue a step-up authentication or place a short precautionary hold autonomously; it may not autonomously file an SAR or close a customer account. A Network Coordination Agent may orchestrate a freeze across multiple banks for amounts and pattern severities within policy thresholds; above those thresholds, a human supervisor approves. The bounded autonomy policy is itself a versioned, auditable artefact maintained by the central authority and the participating institutions jointly. This addresses one of the central concerns raised in BIS Project Noor: explainability and auditability of consequential AI decisions in financial supervision [11].

4.5. Observability and continuous learning

Every agent action emits a structured trace recorded in the observability layer. Traces include the input signals, the agent's reasoning chain (or the model's reason codes), the action taken, and the downstream outcome once labelled. This makes the entire fabric reviewable post-hoc by supervisors, by internal audit, and by external auditors. It also produces a training corpus that the learning group draws on continuously. The Continuous Learning Loop is the architectural mechanism by which the fabric improves between deployments, not only at deployment boundaries.

4.6. Foundation: standards and sovereign primitives

The fabric is built on a foundation of payment messaging standards (ISO 20022 for both domestic and cross-border messages), a tokenisation vault for sensitive identifiers (BVN-equivalent, card PAN, beneficiary name), and security primitives aligned with PCI DSS, ISO/IEC 27001 and sovereign data-residency requirements. The agents operate on tokenised representations; raw sensitive data is decrypted only at explicit, audited de-tokenisation events under legal authority. This preserves the operational realism that has guided the design of first-generation national platforms while enabling the more expansive agent behaviours described above.

5. WORKED EXAMPLE: AGENTIC AI DEPLOYMENT AT THE NIBSS / NPS LAYER

Nigeria offers a particularly suitable substrate for the worked example. The NIBSS National Payment Stack is the youngest of the three reference architectures, with ISO 20022 already native and built-in fraud detection a stated design principle [3]. The CBN's January 2025 directive on settlement-account debiting represents a clear policy mandate to operationalise cross-institution coordination [6]. And the country's PSP / PTSP and agency-banking ecosystem provides a real-world edge-layer environment with intermittent connectivity, where the value of edge-resident agents is most evident.

Figure 3 depicts the end-to-end process flow for a single transaction under the agentic fabric, from initiation through detection, decision, remediation, investigation and learning. The flow assumes the agents described in Section 4 are deployed at the corresponding layers.

Worked Example — Agentic AI Fabric at the NIBSS / NPS Layer (Nigeria)

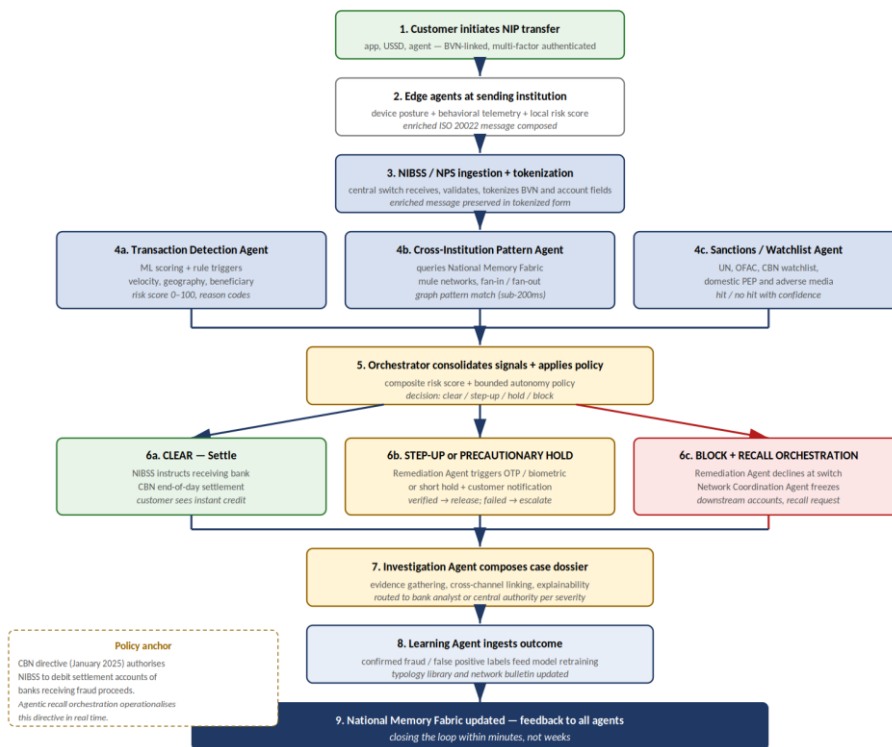


Figure 3 — End-to-end process flow for an agentic AI fabric at the NIBSS / National Payment Stack layer, from transaction initiation through detection, decision, remediation, investigation and continuous learning.

Figure 3. End-to-end process flow for an agentic AI fabric at the NIBSS / National Payment Stack layer.

5.1. Initiation and edge processing

A customer initiates an NIP transfer through their bank app, a USSD channel or an agent terminal. Multi-factor authentication is performed at the sending institution per existing NIBSS requirements. Before the ISO 20022 message is dispatched to the central switch, the edge agents at the sending institution enrich it with device posture, behavioural telemetry and a local risk score. The enrichment preserves user privacy: only tokenised identifiers and aggregated signals are added, not raw biometric or device data.

5.2. Central ingestion and parallel agent invocation

The NIBSS / NPS central switch ingests the enriched message and validates ISO 20022 structure and required fields. The tokenisation service replaces BVN, account number and beneficiary name with tokens, preserving referential integrity so that downstream graph analytics can still link entities. Three agents are then invoked in parallel: a Transaction Detection Agent runs ML scoring and rule triggers on the institutional-layer features; a Cross-Institution Pattern Agent queries the National Memory Fabric for graph patterns involving the sender, the beneficiary or the corridor; a Sanctions / Watchlist Agent checks against UN, OFAC, CBN domestic watchlists and adverse media. The three agents complete within a budget of approximately 200 milliseconds, well within NIP's instant settlement envelope.

5.3. Orchestrator decision and bounded remediation

The Orchestrator consolidates the three agents' outputs into a composite risk profile and applies the bounded autonomy policy. Four outcomes are possible: clear (settle), step-up authentication, precautionary hold, or block with recall orchestration. The cleared path proceeds to receiving-bank credit and end-of-day CBN settlement, unchanged from current NIP operation. The step-up path triggers a Remediation Agent action — typically an OTP, biometric re-authentication, or short hold with customer notification. The block path declines the transaction at the central switch and triggers the Network Coordination Agent to orchestrate downstream freezes if the funds have already been routed to mule accounts in earlier transactions linked to the same beneficiary.

5.4. Investigation and case dossier composition

For any non-clear outcome, the Investigation Agent composes a case dossier in structured form: the original signals, the agents that fired, the orchestrator's reasoning, related transactions across the customer's history and across the beneficiary's network, and a draft narrative. The dossier is routed to the appropriate human reviewer based on severity — bank fraud analyst for routine cases, central authority FIU for material amounts or pattern-level concerns, EFCC or other law enforcement for cases meeting referral thresholds. The dossier dramatically reduces the analyst's preparation time: rather than reconstructing the case from raw logs, the analyst reviews a structured brief.

5.5. Operationalising the CBN January 2025 directive

The CBN directive instructing NIBSS to debit settlement accounts of banks receiving fraud proceeds is the policy anchor for the recall orchestration capability described above. Under the agentic fabric, recall is not a manual after-the-fact process; the Network Coordination Agent can, within policy thresholds, automatically initiate freezes at receiving institutions the moment a confirmed-fraud signal is recorded upstream. The settlement-account debit becomes a final-step enforcement on residual losses, not the primary recovery mechanism. The directive's intent — making receiving banks accountable for detection — is operationalised through real-time coordination rather than ex-post penalty.

5.6. Continuous learning and ecosystem-wide intelligence

Once a case is resolved — confirmed fraud, false positive, or inconclusive — the Learning Agent ingests the outcome and updates the relevant artefacts: the institutional and national models are retrained on the new labelled example; the typology library is updated if a new pattern was identified; a network bulletin is issued to other participating institutions describing the pattern in a manner that preserves customer privacy. The learning cycle closes within minutes for individual outcomes and within days for typology-level updates, replacing the weekly-or-monthly batch retraining of first-generation platforms.

6. BEYOND FRAUD: EXTENSIBILITY OF THE AGENTIC FABRIC

Fraud detection is the natural entry use case for an agentic AI fabric — the latency requirements are stringent, the workflow is well-understood, and the cross-institution coordination need is acute. The same fabric, however, extends to a wider supervisory and operational remit without architectural change.

Anti-Money Laundering and Counter-Financing of Terrorism. The agent role taxonomy already supports AML pattern detection: structuring, layering and integration patterns are graph

patterns that the Cross-Institution Pattern Agent can score continuously. The Investigation Agent composes SAR drafts; the Supervisory Reporting Agent submits to the FIU; the Learning Agent updates AML typology in line with FATF guidance and Project Aurora findings on cross-border AML cooperation [9][17].

KYC and KYB monitoring. Persistent memory at the institutional and national layers turns one-off onboarding KYC into continuous customer due diligence. Behavioural agents detect drift between stated customer profile and observed transactional behaviour; the Investigation Agent assembles change-of-risk-profile dossiers; the Remediation Agent triggers re-KYC where warranted, with bounded autonomy and clear policy gates.

Consumer protection. The fabric supports consumer-protection use cases such as authorised push payment fraud detection, where the customer themselves is socially engineered. Edge behavioural agents can detect coercion signals — unusual transaction velocity, screen-sharing indicators consistent with malware patterns documented in recent threat research [16] — and trigger step-up flows that interrupt the social-engineering script.

Cross-border supervision. Where two or more national fabrics interoperate through ISO 20022 messaging, the National Coordination Agents on either side can negotiate joint actions on cross-border patterns. BIS Project Aurora has already established the technical feasibility of privacy-preserving cross-border AML cooperation [9]; the agentic fabric provides the operational substrate. The Cambridge SupTech Lab's State of SupTech 2024 underscores that cross-border cooperation remains the principal gap in the current supervisory landscape [12], and the architecture proposed here is a deliberate response to that gap.

7. RISKS, LIMITATIONS AND OPEN QUESTIONS

The proposed architecture introduces risks that must be addressed before any operational deployment.

Model and agent governance. Distributing autonomy to agents amplifies the model-risk-management challenge. Every model used by every agent must be governed under a model risk framework: development, validation, monitoring, retirement. BIS Project Noor's emphasis on explainability of high-risk financial AI is directly relevant [11], and the observability layer described in Section 4.5 is the architectural mechanism by which the explainability requirement is met in practice.

Accountability boundaries. When an agent acts, who is accountable for the outcome? Bounded autonomy policies must allocate accountability clearly: institutions for their institutional-layer agents; the central authority for the national-layer agents; vendors for the agents they supply, under contractual model risk obligations. Ambiguity in accountability is the single most likely cause of regulatory rejection.

Regulatory acceptance. Central banks and supervisory authorities are appropriately cautious about autonomous AI in critical financial infrastructure. The proposed architecture mitigates this through bounded autonomy, observability, human-in-the-loop gates above policy thresholds, and explicit alignment with the SupTech direction of BIS Innovation Hub. Pilot deployment within a regulatory sandbox — of the kind several jurisdictions now operate — is the natural path to incremental regulatory acceptance.

Concentration and operational resilience. A national agentic fabric is critical infrastructure. Its operational resilience must meet the standards of the underlying payment system itself: redundancy, failover, business continuity, and graceful degradation. The architecture must also avoid concentration of vendor dependency in agent supply, which would be a single point of systemic risk.

Adversarial adaptation. Fraud adversaries will adapt to an agentic fabric just as they adapt to centralised platforms. The fabric's defence is its continuous learning loop. Whether the loop is fast enough to outpace adversarial adaptation — particularly AI-amplified fraud campaigns documented in recent analysis of PIX-targeting attacks [15][16] — is the central empirical question that pilot deployment must answer.

Open research questions. Several questions are deferred to further research. How is bounded autonomy policy versioning best operationalised across institutions of differing technical maturity? What is the right benchmark for agent-action latency at the national-authority layer? How do agentic patterns interact with privacy-enhancing technologies for cross-border data sharing [9][17]? These questions are tractable within the architectural frame proposed here, but they require empirical pilot data to answer.

8. CONCLUSIONS

First-generation national fraud detection platforms — PIX, UPI, NIBSS — have achieved remarkable scale and speed. Their architectural limits, however, are now visible: human-in-the-loop investigation bottlenecks, batch retraining latency, and weak cross-institution coordination. This paper has proposed an architectural successor in the form of an agentic AI fabric, distributing detection, investigation, remediation and learning across a layered set of cooperating agents with persistent memory, bounded autonomy and continuous observability. A worked example anchored on the NIBSS / National Payment Stack in Nigeria, under the policy direction set by the Central Bank of Nigeria's January 2025 fraud-debit directive, illustrates how the architecture operates end-to-end. The fabric extends naturally to AML, KYC/KYB, consumer protection and cross-border supervision.

The proposition is consistent with the direction set by the BIS Innovation Hub's work on AI in supervision (Project Aurora, Project Noor) and the broader SupTech maturation tracked by the Cambridge SupTech Lab. The risks of model governance, accountability allocation and regulatory acceptance are real and must be addressed through bounded autonomy policy, explicit observability and sandboxed pilot deployment. The empirical question of whether the continuous learning loop can outpace adversarial adaptation is the question pilot deployment must answer.

The contribution of this paper is to bring together two streams — national payment fraud detection and agentic AI — that have evolved separately, and to propose a coherent high-level design that fuses them. The intended audience is the community of central bank technologists, payment system operators and supervisory authorities now considering how to evolve their fraud architectures for the next decade of instant payments.

REFERENCES

- [1] Stripe, (2025) "A guide to Pix payments in Brazil", Stripe Resources. Available at <https://stripe.com/resources/more/pix-replacing-cards-cash-brazil>.
- [2] European Payments Council, (2024) "UPI: revolutionising real-time digital payments in India", Interview with R. Shukla, NPCI International Payments Limited.
- [3] NIBSS, (2025) "National Payment Stack (NPS)", Nigeria Inter-Bank Settlement System Plc. Available at <https://nibss-plc.com.ng/national-payment-stack/>.
- [4] SEON, (2025) "Brazil PIX Incident: Real-Time Payment Fraud Lessons", Incident analysis of the C&M Software cyberattack.
- [5] Billcut, (2025) "New NPCI Alerts Reduce UPI Scams: How the 2025 Update Protects You", UPI safety update commentary.
- [6] TechCabal, (2025) "Exclusive: CBN orders NIBSS to debit banks over fraudulent transactions", Reporting on the January 2025 CBN directive.
- [7] The Paypers, (2026) "Brazil's Central Bank updates Pix security rules", Reporting on precautionary hold and device-trust transfer limits.
- [8] QED Investors, (2025) "The frontlines of fraud: How Brazil is becoming a global testbed for financial crime prevention", Industry commentary on BCB Resolution 589 and MED.
- [9] Bank for International Settlements Innovation Hub, (2023) "Project Aurora: the power of data, technology and collaboration to combat money laundering across institutions and borders", BIS, Nordic Centre.
- [10] Lucinity, (2023) "Charting a New Course Against FinCrime: BIS Innovation Hub and Lucinity Successfully Conclude Project Aurora", Project conclusion commentary.
- [11] Bank for International Settlements Innovation Hub, (2025) "Project Noor: explaining AI models for financial supervision", BIS Hong Kong Centre with HKMA and FCA.
- [12] Cambridge SupTech Lab, (2024) "State of SupTech 2024", Cambridge SupTech Lab global benchmarks.
- [13] Akshansh Jaiswal, (2025) "How UPI Works: A Technical Deep Dive into Its Architecture, Data Flow, and Security", Technical analysis.
- [14] Gochhayat, A., (2025) "The Data Behind UPI: Understanding the Architecture of Real-Time Financial Pipelines".
- [15] ArXiv preprint, (2025) "A Taxonomy of Pix Fraud in Brazil: Attack Methodologies, AI-Driven Amplification, and Defensive Strategies". Available at <https://arxiv.org/pdf/2511.20902>.
- [16] Zimperium, (2026) "PixRevolution: The Agent-Operated Android Trojan Hijacking Brazil's PIX Payments in Real Time", Mobile threat intelligence report.
- [17] Financial Action Task Force, (2023) "FATF Recommendations 11, 16 and 20: Record-keeping, wire transfers and reporting of suspicious transactions", FATF, Paris.
- [18] Bank for International Settlements, (2024) "Payments and Markets Infrastructure: API harmonisation to enhance cross-border payments", BIS Toolkit.
- [19] International Organization for Standardization, (2022) "ISO 20022 Financial Services — Universal financial industry message scheme", ISO, Geneva.
- [20] PCI Security Standards Council, (2024) "PCI Data Security Standard (PCI DSS) Requirements and Security Assessment Procedures", PCI SSC.

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